

1 Questions

1. Compute, with a sentence of why: what is $\sin(210^\circ)$? Explain using the circle picture, not a calculator.
2. Conversions: 135° in radians? And $7\pi/6$ in degrees?
3. Symmetry: list all angles between 0° and 360° where $\sin \theta = -1/2$.
4. Back to the triangle: a right triangle has hypotenuse 13 and an angle of 30° . How long is the side opposite that angle?
5. Wave: the tip takes 8 seconds per full rotation. What is the period and frequency of the height graph - and why doesn't the pattern repeat at 4 seconds, even though the tip is at height zero?
6. Small angle: estimate $\sin(0.02)$ without a calculator. Then say roughly how far off you expect your estimate to be, and why the error is tiny.
7. Proof in miniature: give a one-sentence argument for why $\sin(180^\circ - \theta) = \sin \theta$.
8. Predict, then check: is $\sin(60^\circ)$ bigger or smaller than $1/2$? Roughly how big, by eyeballing the circle? Then check yourself against the exact value.
9. Blank page, next week: from nothing, rebuild the whole arc — rotating tip of a length-1 hypotenuse $\rightarrow$ sine as the height on the unit circle $\rightarrow$ why $\sin \theta \approx \theta$ for small angles. If you can rebuild it cold, you own it.

2 Answers

1. $\sin(210^\circ) = -\frac{1}{2}$
This follows from the pattern of $\sin(30^\circ)$ and $\sin(150^\circ) = \frac{1}{2}$, the key difference being a reflection in the x-axis for the latter and a reflection in both axis on the former, and as we ascertained earlier a reflection in the y-axis for sine has no effect on the output, while a reflection on the x-axis moves into negatives.
2. $135^\circ = \frac{3\pi}{4}$
 $\frac{7\pi}{6} = 210^\circ$
3. Following from question 1, $\sin(210^\circ) = -\frac{1}{2}$
Likewise a reflection in the y-axis shows that $\sin(330^\circ) = -\frac{1}{2}$
4. $\sin(30^\circ) = \frac{O}{13}$
 $13 * \sin(30^\circ) = O$
 $O = 13 * \frac{\pi}{6}$
The opposite side length is $\frac{13\pi}{6}$

5. The period is 8s; it isn't 4s as the gradient facing the opposite direction.
The frequency $f = \frac{1}{T}$ where T is the period.
So $f = \frac{1}{8}$ or $f = \frac{1}{8}Hz$
6. $\sin(0.02) \approx 0.019$
This is because the change in the height near 0 is much greater than the change in width, so the sine function which captures the change in y (height) nearly captures the arc length.
7. $\sin(180^\circ - \theta) = \sin(\theta)$:
starting at 180° and moving backwards θ aligns the y-axis on both sides of the circle. Interestingly adding theta would do the same thing, but swap negative/positive.
8. Looks about 0.8, so bigger than a half.
 $\sin(60^\circ) = \sin(\frac{\pi}{3}) \approx 0.866$